

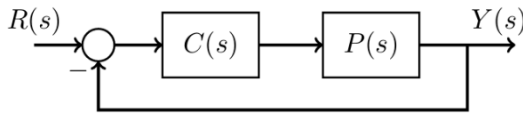
Controls Design Final Project

Introduction

Control systems are essential for ensuring that dynamic systems perform as desired, meeting specific requirements and exhibiting the desired characteristics. Often, these requirements include predefined thresholds or limitations set by an employer or project constraints. In other cases, the focus may be on analyzing a given plant's behavior to stabilize it and achieve optimal performance. In this report, we will explore the process of designing and analyzing a control system for a given plant using the tools and methodologies acquired during our studies. The goal is to ensure the system meets the desired performance criteria while adhering to the given constraints.

Given Case

For the purposes of this report, we will analyze a standard feedback loop, as illustrated in Figure 1. We have an input $R(s)$, a controller $C(s)$, a plant $P(s)$, and an output $Y(s)$. We are considering a set plant behavior defined by Equation (1).



$$P(s) = \frac{s + 10}{(s + 1)^2(s + 2)} \quad (1)$$

Figure 1: Feedback Loop

Analysis

Let's begin by analyzing the system in the absence of a controller, where $C(s) = 1$. This allows the plant $P(s)$ to operate without any external influence, preserving its natural behavior. This situation serves as a baseline for understanding the plant's dynamics and stability under unity feedback. In this case we can calculate our characteristic polynomial as product of numerators plus the product of the denominators in the forward pass $1 * (s + 10) + 1 * (s + 1)^2(s + 2)$. This will simplify to $\Delta(s) = s^3 + 4s^2 + 6s + 12$. In Figure 2 we can see the closed loop poles plotted on the real and Imaginary axis.

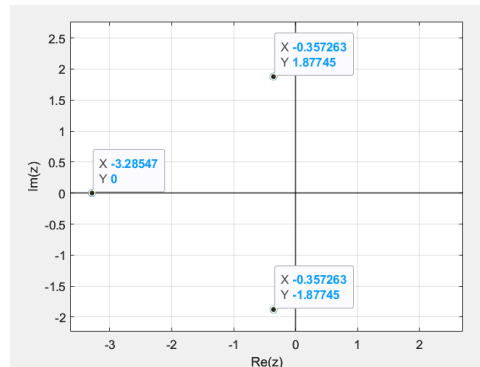


Figure 2: Closed Loop Poles when $C(s) = 1$

Since it is a third order polynomial, we obtained three poles. One negative real and two negative complex poles. From the analysis, we can conclude that the system is stable, as all the poles of the characteristic equation lie in the Open Left Half Plane (OLHP). However, the presence of complex-conjugate poles

introduces oscillatory behavior in the system's response, which may result in undesired oscillations. To address this, we will further explore control strategies to better manage the system's output and improve its performance.

The simplest controller we can design is a Proportional Controller, where the controller is defined as $C(s) = K$ with K being a constant gain. To determine an appropriate value, we use the Routh-Hurwitz Test, which allows us to systematically check whether any roots of the characteristic equation lie in the Closed Right Half Plane (CRHP). Additionally, this test can be applied using variables to identify the range of values that K can take while maintaining stability. For this system, the new characteristic polynomial is derived from the closed-loop characteristic equation:

$$\Delta(s) = s^3 + 4s^2 + (5 + k)s + (2 + 10k)$$

s^3	1	$5 + k$
s^2	4	$2 + 10k$
s^1	$\frac{9 - 3k}{2}$	0
s^0	$2 + 10k$	

There should be no sign changes in the first column, thus we have the inequalities of $2 + 10k > 0$ and $\frac{9-3k}{2} > 0$. After some simplification we concluded that $3 > k > -\frac{1}{5}$ in order to be stable. We can check the Routh-Hurwitz Test using MATLAB for calculating a Root Locus Graph. In cases where we have a scalar parameter the Root Locus comes handy since it shows the movement of the poles as the scalar increases or decreases. Figure 3 shows that we were right with our calculations since we start seeing positive poles when we reach $k = 3$ and in the negative root locus when $k = -0.2$.

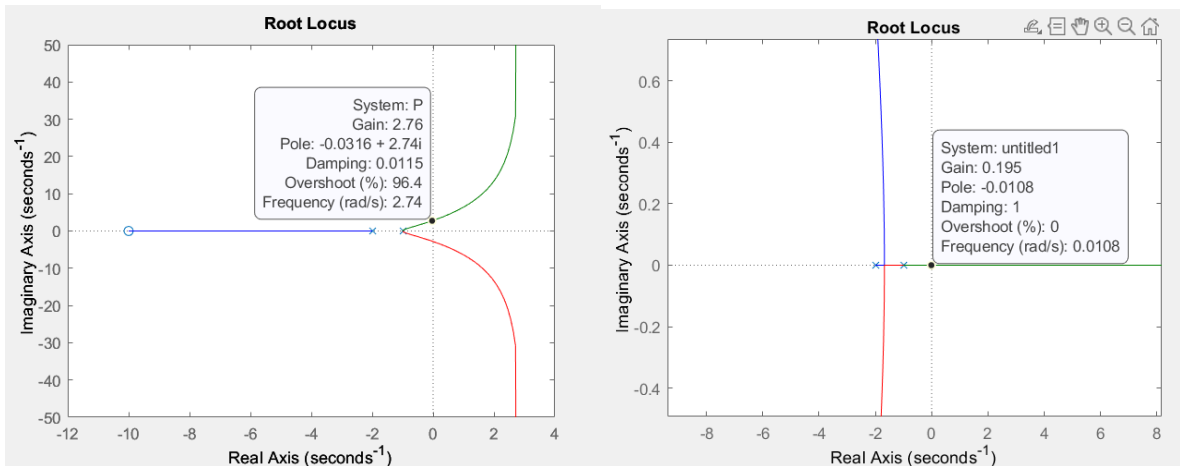


Figure 3: Positive (Left) and Negative (Right) Root Locus of $L(s) = P(s)$

Now that we know the proportional controller can stabilize the system, let's test it with an input to see how well we can approximate the desired output with different signals. A simple input can be a step signal $r(t) = 1$ with $m = 0$ (where $r(t) = t^m$). To calculate the steady state error, we need to define our error output as the input minus the output of the feedback: $E(s) = R(s) - Y(s)$. After simplifying the

equation, replacing the $L(s)$ by its expanded form of $\frac{a(s)}{s^q b(s)}$ (where q is the number of poles at the origin), and applying the Final Value Theorem, we obtain the representative equation for steady state error (2).

$$\lim_{s \rightarrow 0} sE(s) = m! \frac{b(s)}{(s^q b(s) + a(s))s^{m-q}} \quad (2)$$

For our case both q and m are equal to zero. Thus, our Steady State Error is going to be $\lim_{s \rightarrow 0} \frac{b(s)}{b(s)+a(s)}$ giving us an error of $\frac{1}{6}$ or 0.1666.

At this point, we have stability and precision so let's see how far we can take this controller to improve robustness. The Gain Margin (GM) and Phase Margin (PM) are relevant information to accomplish this task. Since our system is stable at $C(s) = 1$ we can say that if we increase a controller $C(s) = k$, the first point in which the poles touch the imaginary axis is at the border of becoming unstable. This happens at $\omega_{cg} = \omega_{cp}$ which are the Cross Over frequencies (frequencies when poles touch the imaginary axis). As we increase our proportional controller, we increase ω_{cg} but ω_{cp} remains since it depends on dynamic controllers. PM and GM are metrics from stable systems that depend on these two frequencies. GM mathematically shows the amplification that K can handle before $\omega_{cg} = \omega_{cp}$ and PM shows how far are we from instability. We can obtain all this information from the bode plot of our plant. Figure 4 shows the plotted Magnitude and Phase Diagrams.

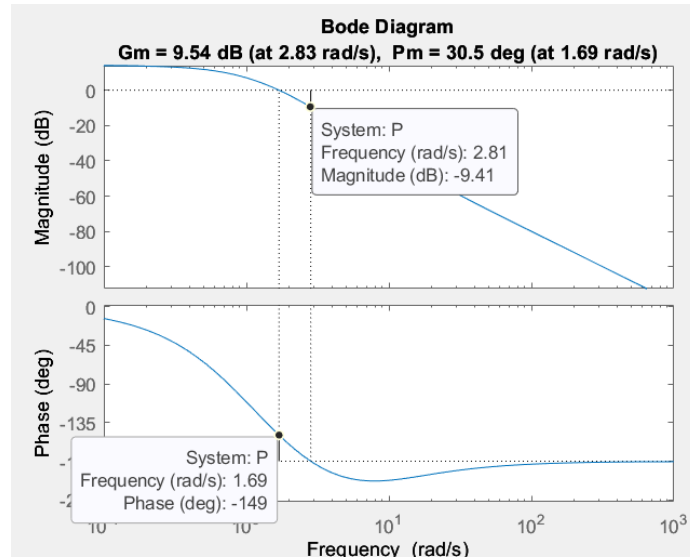


Figure 4: Bode Diagram P(s) when $C(s) = 1$

From the graph analysis, the system with $C(s)=1$ has a GM of 9.54 dB and $\omega_{cp} = 2.83$ rad/s and a PM of 30.5° at $\omega_{cg} = 1.69$ rad/s. These values indicate that the system is moderately stable since it can keep stability even when K increases 3.1 times, but may exhibit oscillatory behavior due to the relatively low PM.

Design of Lag Compensator

Finally, let's design a controller for the same plant $P(s)$ to meet certain specifications. We are looking for a controller that gets an Overshoot (Os) of 25% at most with a unit step and a Tracking Error less than 0.1 with a unit step.

To begin with the design let's define the form of the Lag Compensator Controller. As shown in equation (3), these types of controllers have both a proportional and a dynamic component determined by a gain K_c and a bode plot form of a zero and a pole. In this case the lag compensator has its zero greater than its pole, so we replaced the zero by a pole multiplied by a constant β that is greater than one.

$$C(s) = K_c \beta \frac{\frac{s}{\beta p} + 1}{\frac{s}{p} + 1} \quad (3)$$

As we can assume and see in Figure 5 the plots of this controller have an amplification at lower frequencies and a phase decrease between $f_p = p$ and $f_z = \beta p$. The theory behind this controller is that since the phase sum can only hurt our ω_{cg} , we can compensate for it by changing our magnitude at low frequencies (highly related to the tracking error of our system).

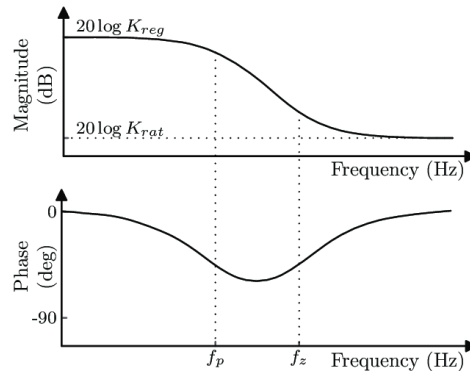


Figure 5: Bode Plot of a Lag Compensator

This means that we must begin our design by meeting our desired PM (related by equation (4) to the Os) and later correct the tracking error of the system with equation (2).

$$e^{\frac{\pi \rho}{\sqrt{1-\rho^2}}} = os \quad \text{and} \quad PM = 100\rho \quad (4)$$

$$e^{\frac{\pi \rho}{\sqrt{1-\rho^2}}} = 0.25 \quad \rightarrow \quad \rho = 0.404 \quad \text{and} \quad PM = 40$$

Using those equations, we calculated the desired phase margin to be 40. Since there is going to be a shift on ω_{cg} we need to buffer that PM by adding around 10 degrees so our desired PM = 50. Now we need to see at which frequency we can obtain that Phase Margin that should be at around 130 degrees (50 from 180). From Figure 4 we can see that it happens at a frequency of 1.28 rad/s. In order to get that PM we will need to shift down our magnitude plot at that frequency so that $\omega_{cg} = 1.28$ rad/s. The same Figure shows that at that frequency, our magnitude is 4.14 db. Equation (5) shows mathematically how the shift down relates with K_c .

$$20 \log |K_c| = -db \text{ shift} \rightarrow K_c = 0.621$$

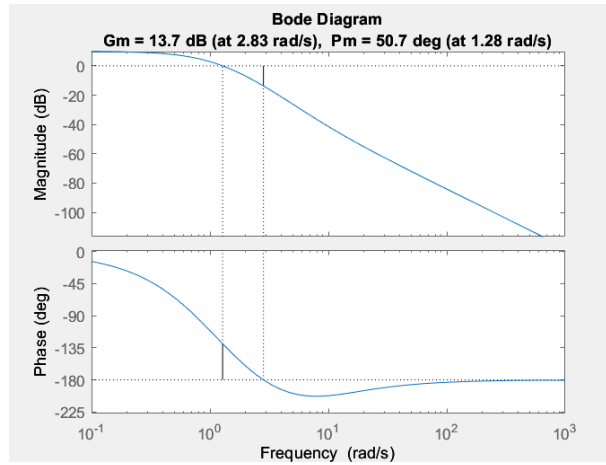


Figure 6: Bode Diagram of C(s)P(s) when K = 0.621

Now that we have obtained the desired PM, we need to meet our specification for steady state error. We can calculate the desired steady state error if we now apply the formula with our plant and our new controller. This will allow us to set an error and get our ratio of pole to zero.

$$sE(s) = \frac{s}{s} \frac{1}{1 + \frac{s+10}{(s+1)^2(s+2)} * K_c \beta \frac{\frac{s}{\beta p} + 1}{\frac{s}{p} + 1}} = \frac{1}{1 + 5K_c \beta}$$

$$0.1 = \frac{1}{1 + 5 * 0.621 * \beta} \rightarrow \beta = 2.8986$$

Finally, we need to compensate for the damage we caused on the phase. This is accomplished by shifting the negative hump of the controller frequency, so it happens one decade before ω_{cg} of KP(s), so it does not influence. This negative hump is related to the position of zero of our controllers so we will define the zero at one decade to the left.

$$\text{with } z = 0.128 \quad p = \frac{z}{\beta} \rightarrow p = 0.04415$$

$$C(s) = (0.621)(2.8986) \frac{\frac{s}{(0.04415)(2.8986)} + 1}{\frac{s}{(0.04415)} + 1} \quad (5)$$

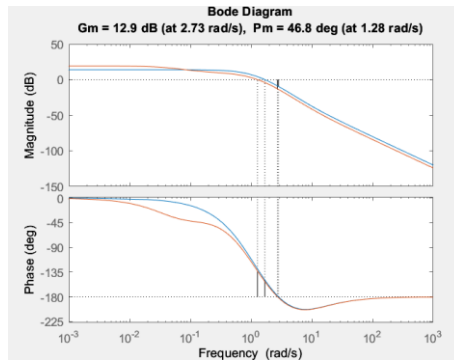


Figure 7: Bode Plot of P(s) vs C(s)

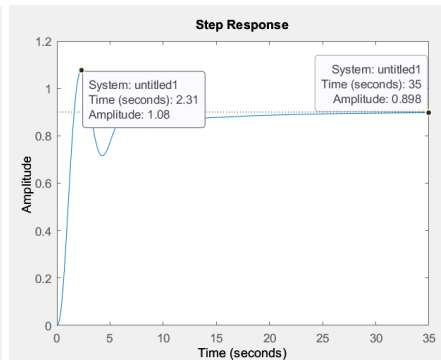


Figure 8: Step Response of H(s)

Our final controller equation will be described by equation (5). However, we can observe in Figure 7 and 8 that although we achieved our Os specification, our error specification is a bit off for the controller. We can easily solve it by lowering our expected tracking error from 0.1 to 0.09. After recalculating the equation our final controller is expressed in equation (6).

$$C(s) = (0.621)(3.25639) \frac{\frac{s}{(0.0393)(3.25639)} + 1}{\frac{s}{(0.0393)} + 1} \quad (6)$$

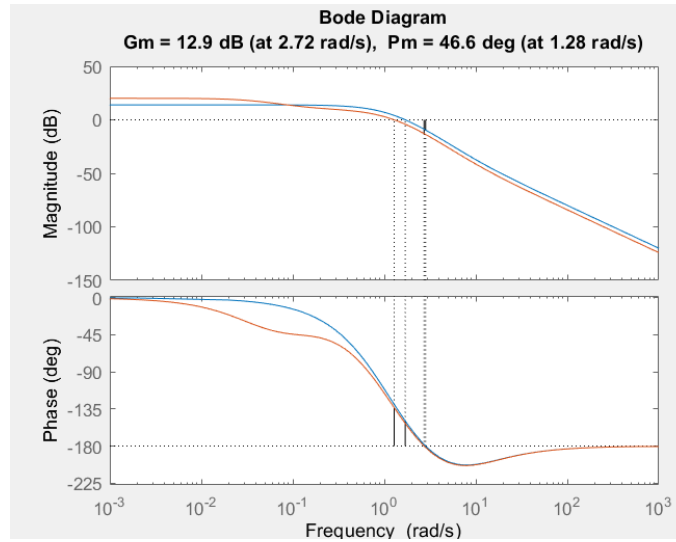


Figure 9: Bode Diagram for new C(s) vs P(s)

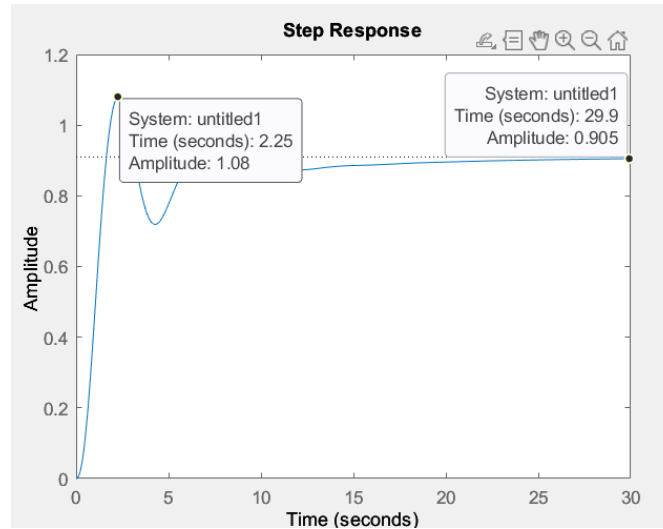


Figure 9: Step Response for new H(s)

Conclusion

In this report, we explored the process of designing and analyzing a control system for a given plant, starting with a baseline analysis under unity feedback and progressing toward the implementation of a lag compensator to meet specific performance criteria. Initially, we determined that the system was stable for a proportional controller with $C(s)=K$ in the range $-\frac{1}{5} < K < 3$, but it exhibited oscillatory behavior due to

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complex-conjugate poles. To address this, we designed a lag compensator to improve phase margin and achieve a maximum overshoot of 25% with a tracking error below 0.1.

Through iterative calculations and analysis using tools such as the Bode plot, root locus, and steady-state error formulas, we successfully derived the compensator parameters to meet these specifications. However, adjustments were necessary to refine the tracking error, resulting in the final controller equation in (6). The final design achieved the desired stability, precision, and robustness, as confirmed by the improved step response and Bode diagram. This demonstrates the effectiveness of the systematic control design process in achieving desired performance while adhering to practical constraints.